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**B. Tech. Degree VI Semester Special Supplementary Examination
November 2014**

**ME 606 CAD/CAM
(2006 Scheme)**

Time: 3 Hours

Maximum Marks: 100

**PART A
(Answer ALL questions)**

(8 x 5 = 40)

- I. (a) Distinguish wire frame modeling from solid frame modeling.
(b) Describe the engineering analysis using FEM.
(c) What are the advantages of numerical control?
(d) Write a note on analog displacement measurement system.
(e) What are the sources of static and dynamic errors in CNC?
(f) What are the features of machining centres?
(g) Mention the applications of FMS.
(h) What are the common languages used in robotic programming?

PART B

(4 x 15 = 60)

- II. What is the line balancing? Describe any three line balancing algorithms, with a suitable example.

OR

- III. (a) Explain the data exchange between CAD and CAM.
(b) Explain the transfer machines used in automation, with neat sketches.

- IV. Explain the following.

- (i) Open and closed loop system
(ii) Position control in CNC
(iii) Linear and rotary displacement transducer

OR

- V. (a) What is CAPP? Explain the major CAPP languages and its structure.
(b) Write a note on canned cycles.

- VI. Explain the automatic tool changer and automatic pallet changer, with neat sketches.

OR

- VII. (a) What are the special design features in CNC machine tools?
(b) Write a note on testing of NC machine tools.

- VIII. (a) Explain the classification of robots.
(b) Describe the control systems in robots.

OR

- IX. (a) Describe the direct numerical control.
(b) Write a note on expert systems.
